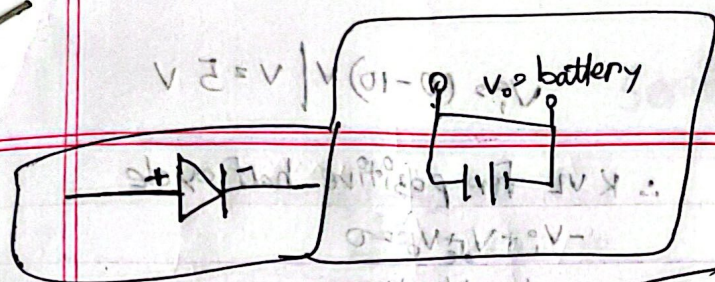


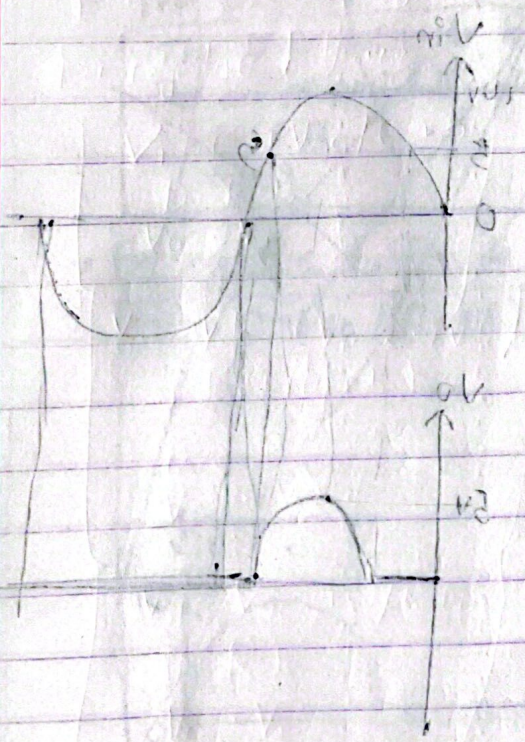
Clippers



Series & KVL

- Parallel & Diode এর দুই দিক
- দুই দিক F.B. হইলে F.B. R.B. হইলে R.B.
- দুই দিক আলোদ হইলে, যা R Power হইবে

- Parallel Forward পাঠিল, $V_o = V_{dc}$, battery polarity রাখা রাখত হবে
- Parallel Reverse পাঠিল, $V_o = V_{in}$, polarity রাখা রাখত হবে



graph must be with pensil & ruler

circuit with pen

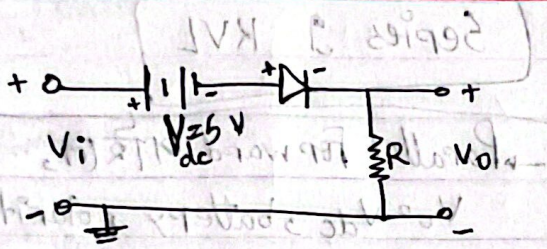
উ পাঠ graph এর $V_o = V_{in}$
নিচের graph এর $V_o = V_o$

$V_i = V_m \sin \omega t$
 $V_o = V_m \sin \omega t$
Diode R.B.
 $V_o = 0$

For negative half cycle the diode will be reversed biased and the output will be zero.

Series

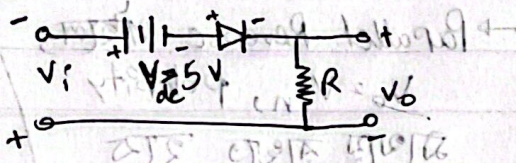
$$V_o = (0-10)V \mid V = 5V$$



∴ KVL for positive half cycle

$$-V_i + V_{dc} + V_o = 0$$

$$V_o = V_i - V_{dc}$$



∴ KVL for negative half cycle

$$+V_i + V_{dc} + V_o = 0$$

$$V_o = -V_i - V_{dc}$$

∴ For positive half cycle:

$$V_i = 0 \rightarrow V_{dc} = (0-5)V$$

Diode will be R.B

V_o will be 0V

$$V_i = 0 \rightarrow V_{dc} \rightarrow V_{max} \rightarrow V_{dc}$$

$$V_i = (5 \rightarrow 10 \rightarrow 5)V$$

$$V_i = 6 \mid V_o = 1V \mid D = F.B$$

$$V_i = 7 \mid V_o = 2V$$

$$V_i = 8 \mid V_o = 3V$$

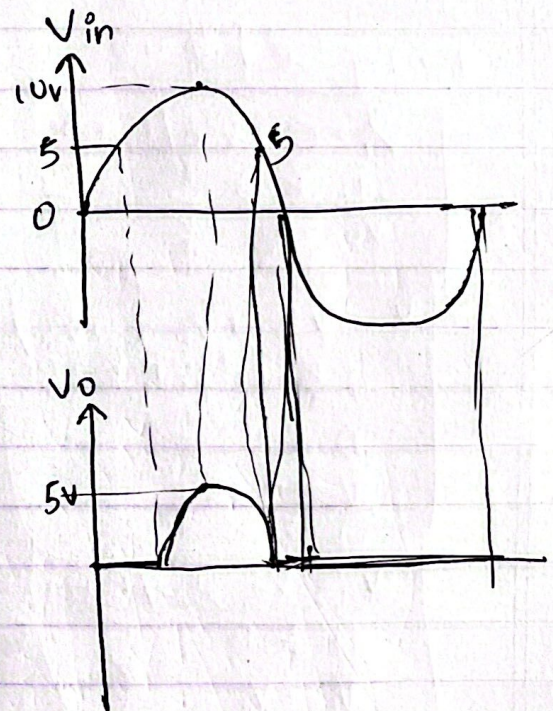
$$V_i = 9 \mid V_o = 4V$$

$$V_i = 10 \mid V_o = 5V$$

$$V_i = V_{dc} \rightarrow 0/5 \rightarrow 0$$

Diode is R.B

$V_o = 0V$



∴ For negative half cycle

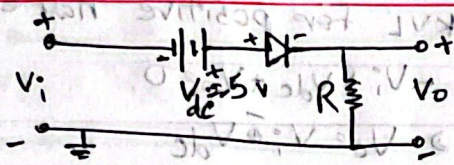
The Diode will be reversed bias

for entire negative half cycle,

thus open,

$$V_o = 0V \mid D = R.B$$

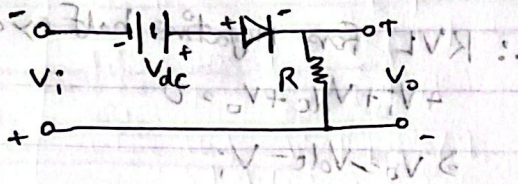
Series



∴ KVL for Positive half cycle

$$-V_i - V_{dc} + V_o = 0$$

$$\Rightarrow V_o = V_i + V_{dc}$$



∴ KVL for negative half cycle

$$+V_i - V_{dc} + V_o = 0$$

$$\Rightarrow V_o = V_{dc} - V_i$$

∴ For positive half cycle

The Diode will be R/B for entire pos. half cycle
 $D = R \cdot B$
 $V_o = 0$

∴ For negative half cycle

$$V_i = 0 \rightarrow V_{dc} / (0 \rightarrow 5) V$$

∴ For positive half cycle

$V_i = 0$	$V_o = 5$
$V_i = 1$	$V_o = 6$
$V_i = 2$	$V_o = 7$
$V_i = 3$	$V_o = 8$
$V_i = 4$	$V_o = 9$
$V_i = 5$	$V_o = 10$
$V_i = 6$	$V_o = 11$
$V_i = 7$	$V_o = 12$
$V_i = 8$	$V_o = 13$
$V_i = 9$	$V_o = 14$
$V_i = 10$	$V_o = 15$

$$D = R \cdot B$$

∴ For negative half cycle

$$V_i = 0 \rightarrow V_{dc} / (0 \rightarrow 5) V$$

$$V_o = 0 / D = R \cdot B$$

$$V_i = 0 \rightarrow V_{dc} / (0 \rightarrow 5) V$$

$$D = R \cdot B$$

$$V_i = 0 / V_o = 5$$

$$V_i = 1 / V_o = 4$$

$$V_i = 2 / V_o = 3$$

$$V_i = 3 / V_o = 2$$

$$V_i = 4 / V_o = 1$$

$$V_i = 5 / V_o = 0$$

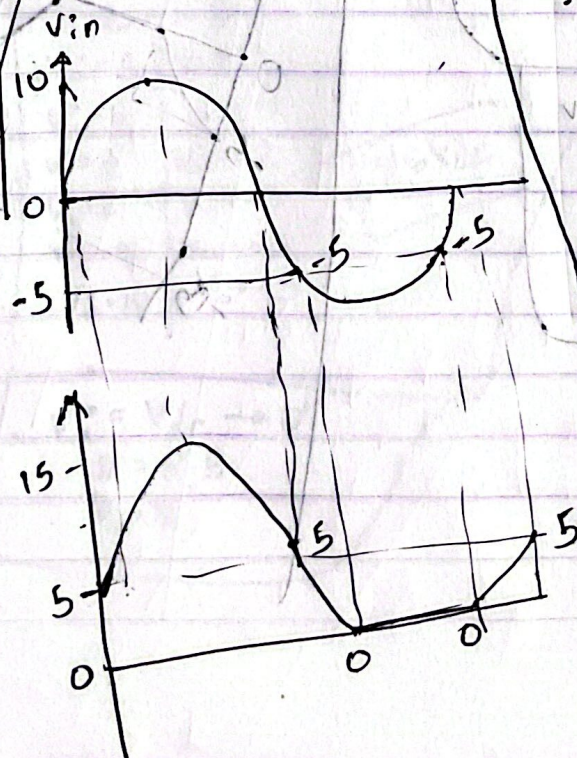
$$V_i = V_{dc} \rightarrow 0 / 5 \rightarrow 0 V$$

same
 $D = R \cdot B$
 $V_o = \text{same}$

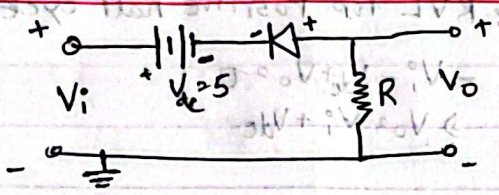
$$V_i = V_{dc} \rightarrow V_{max} \rightarrow V_{dc} / 5 \rightarrow 10 \rightarrow 5 V$$

$$D = R \cdot B$$

$$V_o = 0$$



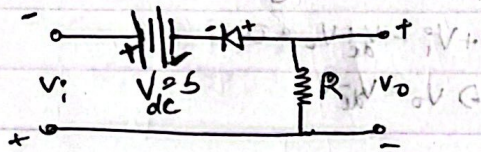
Series



∴ KVL for positive half cycle

$$-V_i + V_{dc} + V_o = 0$$

$$\Rightarrow V_o = V_i - V_{dc}$$



∴ KVL for negative half cycle

$$+V_i + V_{dc} + V_o = 0$$

$$\Rightarrow V_o = -V_{dc} - V_i$$

∴ For positive half cycle:

$$V_i = 0 \rightarrow V_o = 0 - 5 = -5V$$

$$V_i = 0 \mid V_o = -5V$$

$$V_i = 1 \mid V_o = -4V$$

$$V_i = 2 \mid V_o = -3V$$

$$V_i = 3 \mid V_o = -2V$$

$$V_i = 4 \mid V_o = -1V$$

$$V_i = 5 \mid V_o = 0V$$

$D = F.B$

$$V_i = 0 \rightarrow V_{dc} \rightarrow V_{max} \rightarrow V_{dc} / 5 \rightarrow 10 \rightarrow 5$$

$$D = R.B$$

$$V_o = 0V$$

$$V_i = V_{dc} \rightarrow 0 / 5 \rightarrow 0V$$

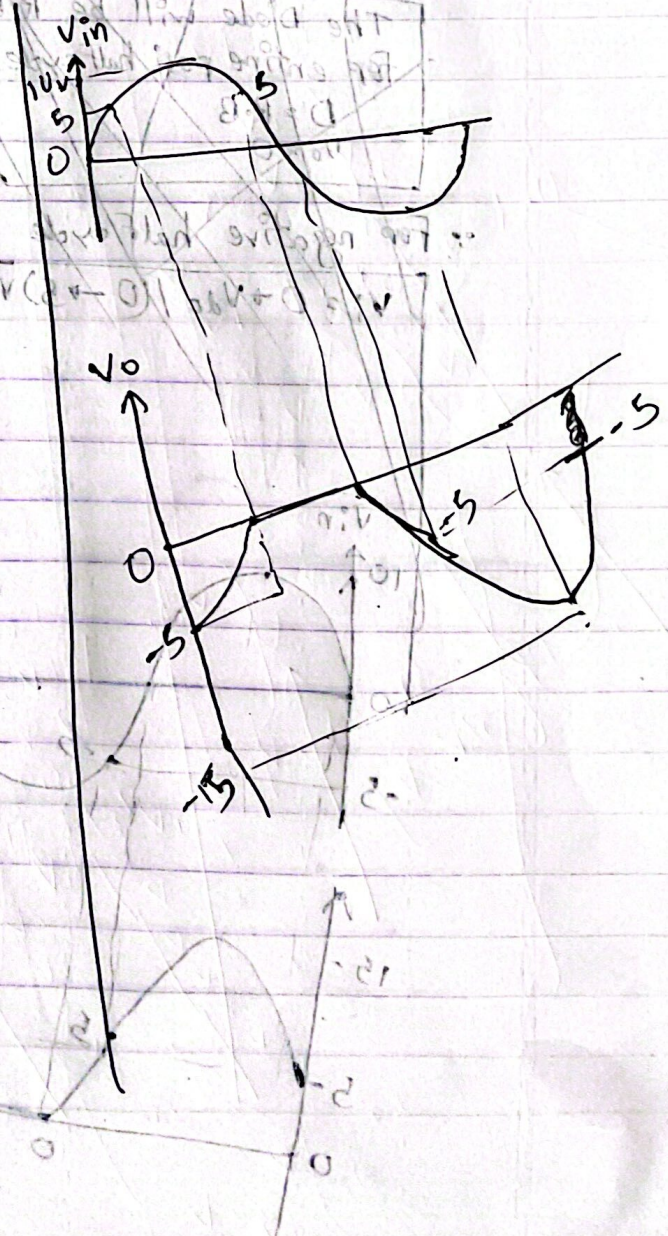
$$D = F.B$$

$$V_o = \text{same}$$

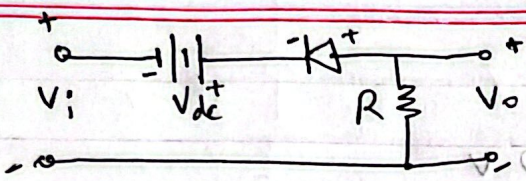
∴ For negative half cycle

$$D = F.B$$

$$V_o = -(V_i + V_{dc})$$



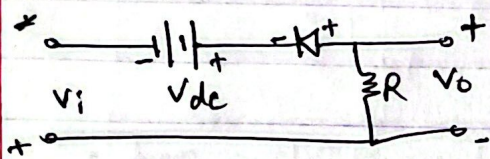
Series



∴ KVL for positive half cycle

$$-V_i - V_{dc} + V_o = 0$$

$$\Rightarrow V_o = V_i + V_{dc}$$



∴ KVL for negative half cycle

$$+V_i - V_{dc} + V_o = 0$$

$$\Rightarrow V_o = V_{dc} - V_i$$

∴ For positive half cycle:

$$D = R \cdot B$$

$$V_o = 0 \text{ V}$$

∴ For negative half cycle:

$$V_i = 0 \rightarrow V_{dc} / 0 \rightarrow 5 \text{ V}$$

$$D = R \cdot B$$

$$V_o = 0$$

$$V_i = V_{dc} \rightarrow V_{max} \rightarrow V_{dc} / 5 \rightarrow 10 \rightarrow 5$$

$$D = R \cdot B$$

$$V_i = 5 \quad V_o = 0$$

$$V_i = 6 \quad V_o = 1$$

$$V_i = 7 \quad V_o = 2$$

$$V_i = 8 \quad V_o = 3$$

$$V_i = 9 \quad V_o = 4$$

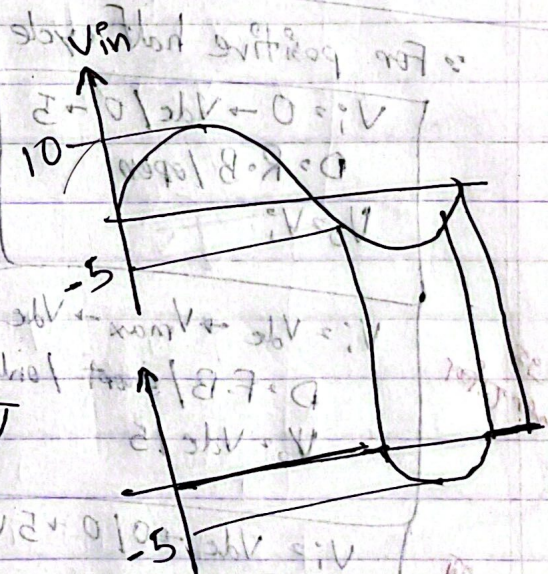
$$V_i = 10 \quad V_o = 5$$

$$V_i = V_{dc}$$

$$V_i = V_{dc} \rightarrow 0$$

$$D = R \cdot B$$

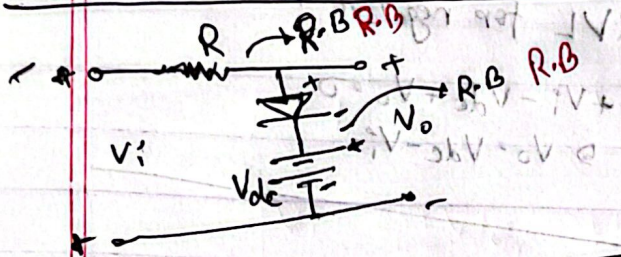
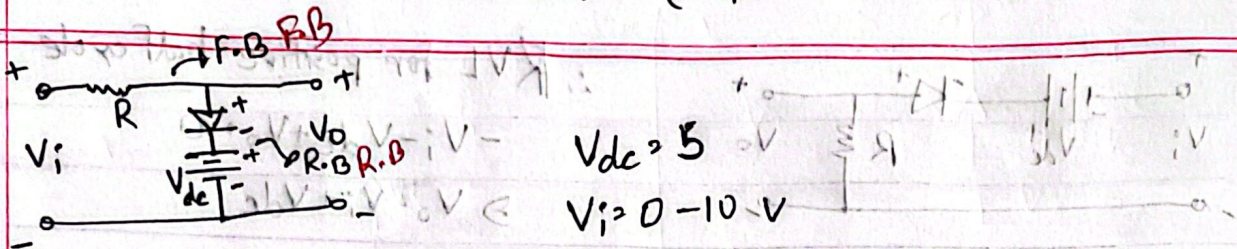
$$V_o = 0$$



∴ For negative half cycle

$$D = R \cdot B$$

Parallel (চার সক্রিয় বেশি)



For positive half cycle

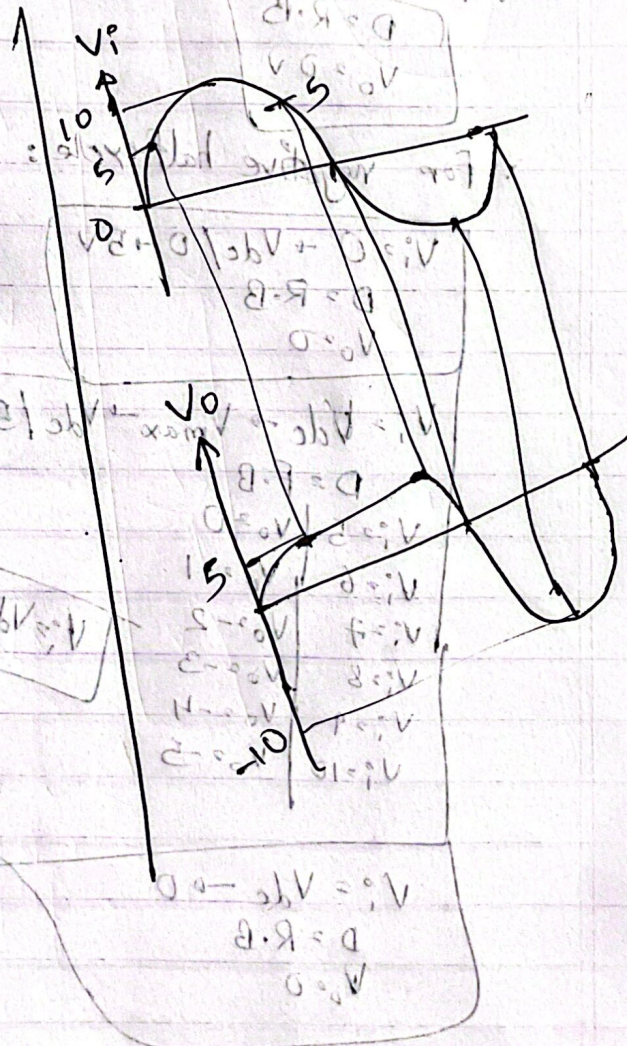
$V_i = 0 \rightarrow V_{dc} / 0 \rightarrow 5$
 $D = R.B / \text{open}$
 $V_o = V_i$

$V_i = V_{dc} \rightarrow V_{max} \rightarrow V_{dc}$
 $D = F.B / \text{short} / \text{only battery}$
 $V_o = V_{dc} / 5$

$V_i = V_{dc} \rightarrow 0 / 0 \rightarrow 5 \text{ V}$
 $D = R.B / \text{open}$
 $V_o = V_i$

For negative half cycle

$D = R.B / \text{open}$
 $V_o = V_i$



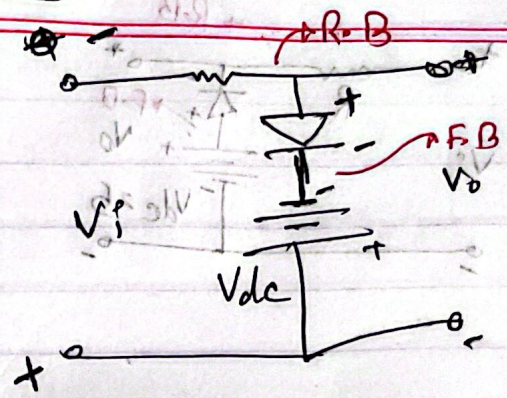
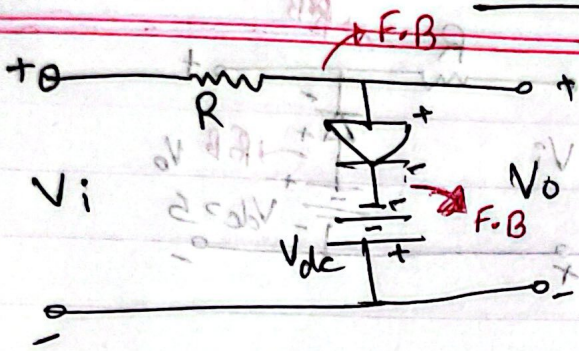
F.B ১ম power

F.B ২য় power

$V_i = 5$ RB
 $V_i = 0$ RB

Parallel

(7)



∴ For positive half cycle:

$$D = F.B / \text{open short}$$

$$V_o = V_{dc}$$

∴ For negative half cycle:

$$V_i = 0 \rightarrow V_{dc} / 0 \rightarrow 5$$

$$D = R.B$$

$$V_o = -V_{dc}$$

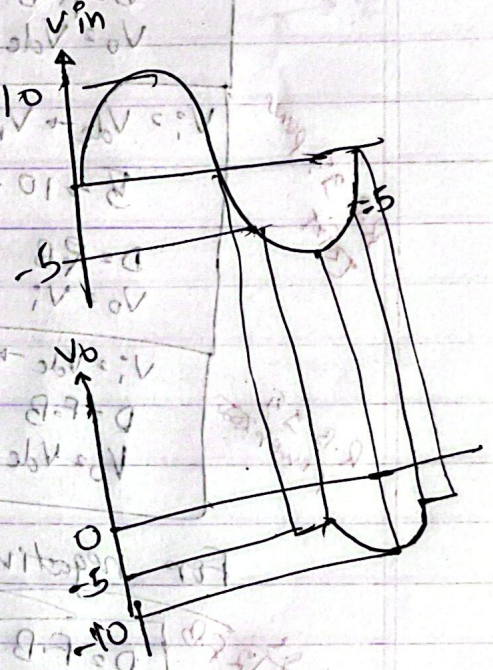
$$V_i = V_{dc} \rightarrow V_{max} \rightarrow V_{dc} / 5 \rightarrow 10 \rightarrow 5$$

$$V_o = V_i / D = R.B$$

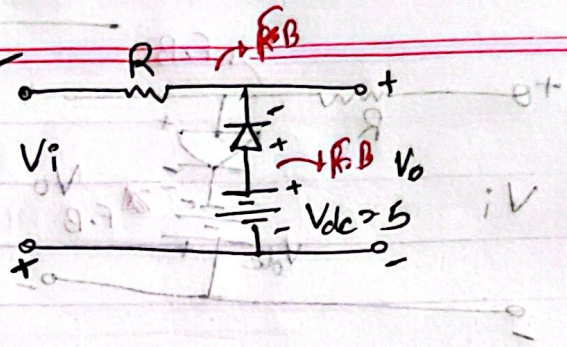
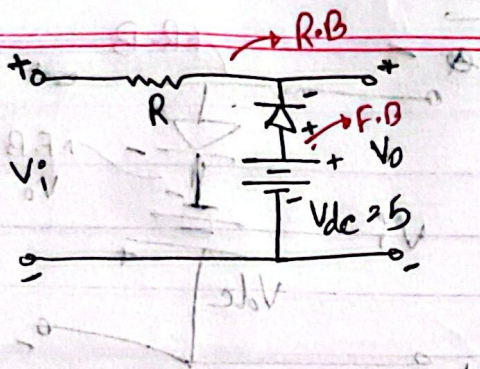
$$V_i \neq V_{dc} \rightarrow V_o$$

$$D = F.B$$

$$V_o = V_i - V_{dc}$$



Parallel



For positive half cycle,

R.B 3/4 power, 1/4 F.B

$$V_i = 0 \rightarrow V_{dc} / 0 \rightarrow 5$$

$$D = R.B$$

$$V_o = V_{dc} / 5$$

$$V_i = V_{dc} \rightarrow V_{max} \rightarrow V_{dc} / 5$$

$$D = R.B$$

$$V_o = V_i$$

$$V_i = V_{dc} \rightarrow 0 / 5 \rightarrow 0$$

$$D = F.B$$

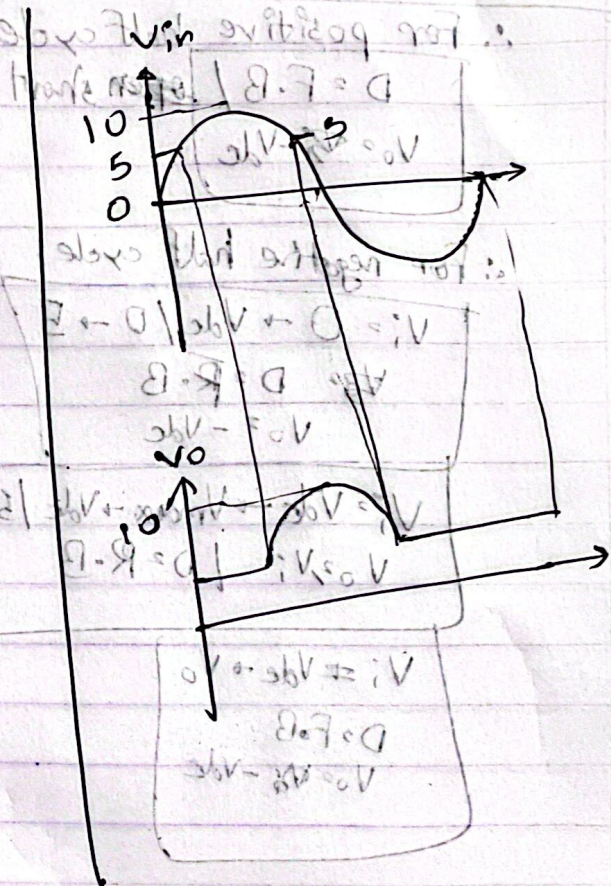
$$V_o = V_{dc} / 5$$

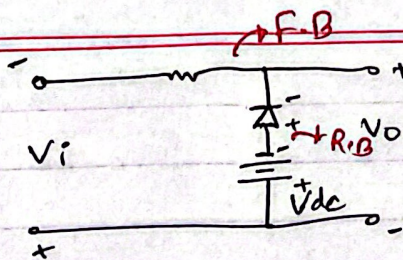
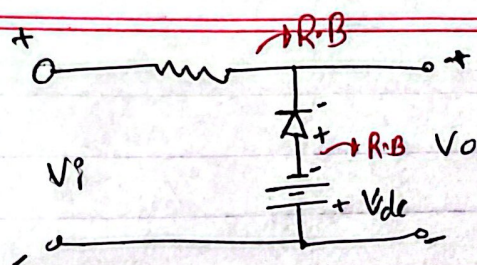
For negative half cycle:

1/4 R.B, 3/4 F.B

$$D = F.B$$

$$V_o = V_{dc}$$





For positive half cycle :

$$D = R \cdot B / \text{open}$$

$$V_o = V_i$$

For negative half cycle :

$$V_i = 0 \rightarrow V_{dc} / 10 \rightarrow 5$$

$$D = R \cdot B$$

$$V_o = -V_i$$

$$V_i = V_{dc} \rightarrow V_{max} \rightarrow V_{dc} / 5 \rightarrow 10 \rightarrow 5$$

$$D = F \cdot B$$

$$V_o = -V_{dc}$$

$$V_i = V_{dc} \rightarrow 0 / 5 \rightarrow 0$$

$$D = R \cdot B$$

$$V_o = -V_i$$

